

WEEK 3 MANDATORY HANDS ON

Spring Rest:

Exersice 1: Create a Spring Web Project using Maven

SpringLearnApplication.java

```
package com.cognizant.spring_learn;

import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication public class
SpringLearnApplication {

    public static void main(String[] args) {
        SpringApplication.run(SpringLearnApplication.class, args);
    }
}
```

1. Create a REST Controller

[src/main/java/com/cognizant/springlearn/HelloController.java](#)

HelloController.java

```
package com.cognizant.spring_learn;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController

public class HelloController {

    @GetMapping("/hello")
```

```
        public String sayHello() {  
return "Hello from Spring Boot!";  
        }  
    }  
}
```

Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?> <project  
xmlns="http://maven.apache.org/POM/4.0.0"  
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
    <modelVersion>4.0.0</modelVersion>
```

```
    <parent>  
        <groupId>org.springframework.boot</groupId>  
        <artifactId>spring-boot-starter-parent</artifactId>  
        <version>3.5.3</version>  
        <relativePath/>  
    </parent>
```

```
    <groupId>com.cognizant</groupId>  
    <artifactId>spring-learn</artifactId>  
    <version>0.0.1-SNAPSHOT</version>  
    <name>spring-learn</name>  
    <description>Demo project for Spring Boot</description>
```

```
    <properties>  
        <java.version>21</java.version>  
    </properties>
```

```
    <dependencies>
```

```
        <dependency>  
            <groupId>org.springframework.boot</groupId>  
            <artifactId>spring-boot-starter-web</artifactId>  
        </dependency>
```

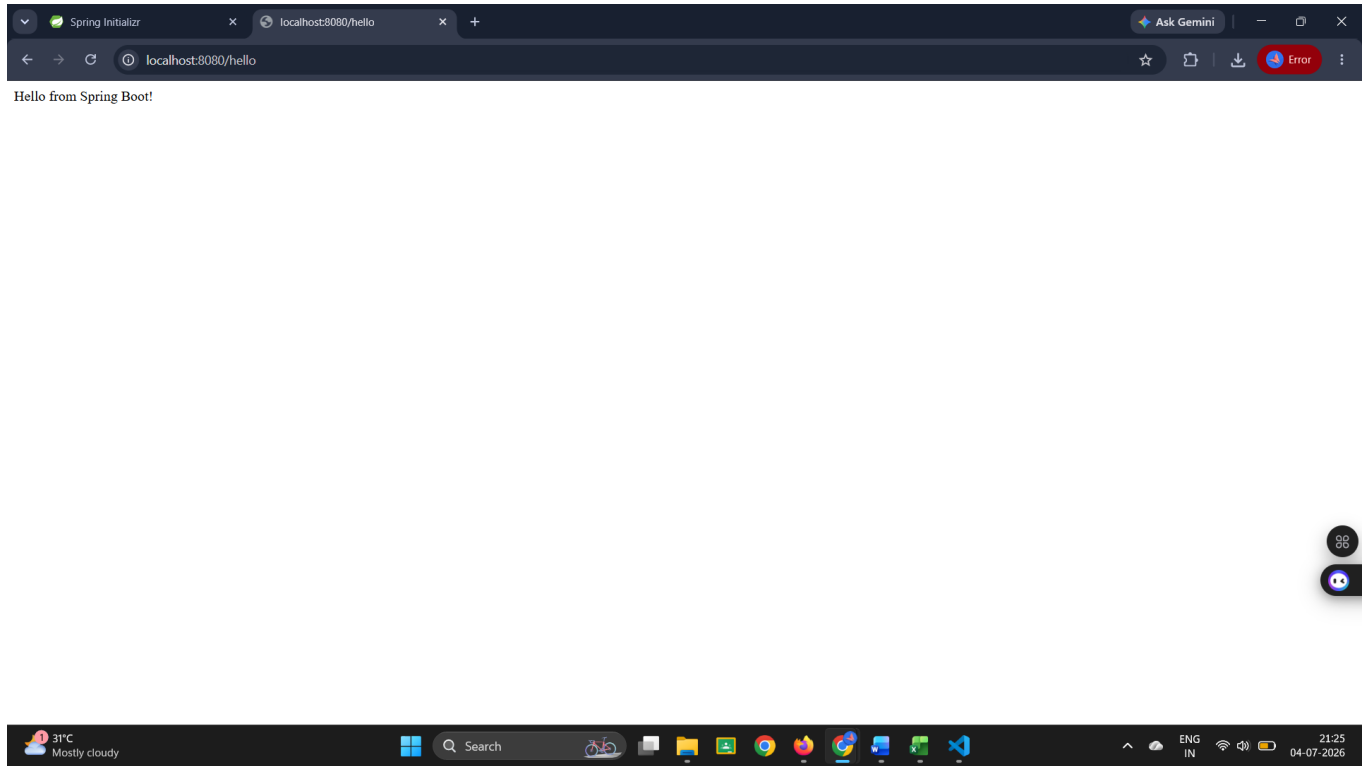
```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-devtools</artifactId>
  <scope>runtime</scope>
  <optional>true</optional>
</dependency>
```

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>
```

```
<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
      <version>3.5.3</version>
    </plugin>
  </plugins>
</build>
```

```
</project>
```

OUTPUT:



Exersice 2:Spring Core – Load Country from Spring Configuration XML

Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<parent>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-parent</artifactId>
```

```
<version>3.5.3</version>
```

```
<relativePath/> <!-- lookup parent from repository -->
```

```
</parent>
```

```
<groupId>com.cognizant</groupId>
```

```
<artifactId>spring-learn</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>
```

```
<name>spring-learn</name>
```

```
<description>Demo project for Spring Boot</description>
```

```
<properties>
```

```
<java.version>21</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
<!-- Spring Boot Web Starter -->
```

```
<dependency>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<!-- Spring Boot DevTools -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-devtools</artifactId>
  <scope>runtime</scope>
  <optional>true</optional>
</dependency>

<!-- SLF4J Logging -->
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>2.0.13</version>
</dependency>

<!-- Testing -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
```

</project>

Java Class:

Country.java:

```
package com.cognizant.spring_learn;

import org.slf4j.Logger; import
org.slf4j.LoggerFactory;

public class Country {    private static final Logger LOGGER =
LoggerFactory.getLogger(Country.class);    private String code;    private String
name;

    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() {
        LOGGER.debug("Inside getCode");        return
code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode");        this.code
= code;
    }

    public String getName() {
        LOGGER.debug("Inside getName");        return
name;
    }

    public void setName(String name) {
        LOGGER.debug("Inside setName");        this.name
= name;
    }
}
```

```

    @Override    public String toString() {        return "Country
[code=" + code + ", name=" + name + "];
    }
}

```

SpringLearnApplication:

```

package com.cognizant.spring_learn;

import org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {

    public static void main(String[] args) {
System.out.println("START");    displayCountry();
        System.out.println("END");
    }

    public static void displayCountry() {
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = (Country) context.getBean("country", Country.class);
System.out.println("Country: " + country.toString());
    } }

```

Country.xml:

```

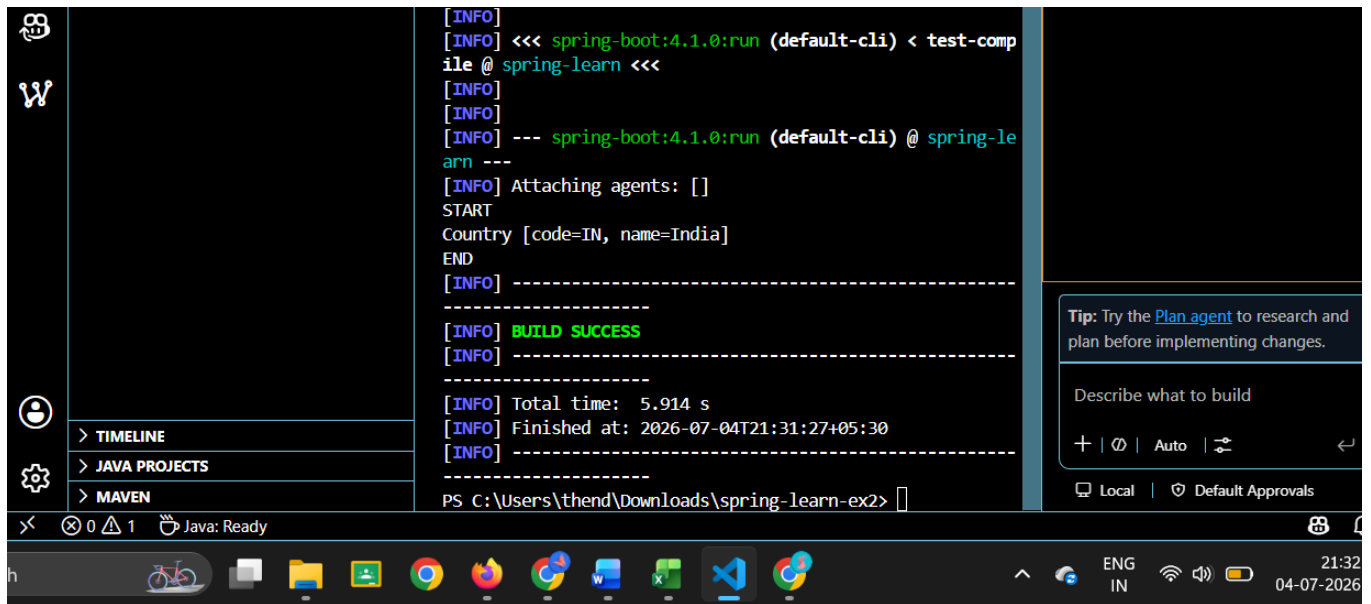
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="country" class="com.cognizant.spring_learn.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean>

</beans>

```


OUTPUT:



```
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-comp
[INFO] file @ spring-learn <<<
[INFO]
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ spring-le
[INFO] arn ---
[INFO] Attaching agents: []
START
Country [code=IN, name=India]
END
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.914 s
[INFO] Finished at: 2026-07-04T21:31:27+05:30
[INFO] -----
PS C:\Users\tend\Downloads\spring-learn-ex2>
```

Exersice 3: Helloworld RESTFUL webservice Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?> <project
xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
```

```
<modelVersion>4.0.0</modelVersion>
```

```
<parent>
```

```
<groupId>org.springframework.boot</groupId>
```

```
<artifactId>spring-boot-starter-parent</artifactId>
```

```
<version>3.5.3</version>
```

```
<relativePath/> <!-- lookup parent from repository -->
```

```
</parent>
```

```
<groupId>com.cognizant</groupId>
```

```
<artifactId>spring-learn</artifactId>
```

```
<version>0.0.1-SNAPSHOT</version>
```

```
<name>spring-learn</name>
```

```
<description>Hello World RESTful Web Service Project</description>
```

```
<properties>
```

```
  <java.version>21</java.version>
```

```
</properties>
```

```
<dependencies>
```

```
  <!-- Spring Boot Web Starter -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-web</artifactId>
```

```
  </dependency>
```

```
  <!-- Spring Boot DevTools (for hot reload during development) -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-devtools</artifactId>
```

```
    <scope>runtime</scope>
```

```
    <optional>true</optional>
```

```
  </dependency>
```

```
  <!-- Testing (optional) -->
```

```
  <dependency>
```

```
    <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-test</artifactId>
```

```
    <scope>test</scope>
```

```
  </dependency>
```

```
</dependencies>
```

```
<build>
```

```
  <plugins>
```

```
    <!-- Spring Boot Maven Plugin -->
```

```
    <plugin>
```

```
      <groupId>org.springframework.boot</groupId>
```

```
      <artifactId>spring-boot-maven-plugin</artifactId>
```

```
      <version>3.5.3</version>
```

```
    </plugin>
```

```
  </plugins>
```

```
</build> </project>
```

Java Class:

HelloController.java

```
package com.cognizant.Spirng_learner.controller;

import org.slf4j.Logger; import org.slf4j.LoggerFactory; import
org.springframework.web.bind.annotation.GetMapping; import
org.springframework.web.bind.annotation.RestController;

@RestController public class
HelloController {

    private static final Logger LOGGER =
LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.debug("START");
        LOGGER.debug("END");
        return "Hello World!!";
    } }
```

SpringlearnerApplication.java

```
package com.cognizant.Spirng_learner;

import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication public class
SpringlearnerApplication {    public
static void main(String[] args) {
```

```
    SpringApplication.run(SpringlearnerApplication.class, args);  
  }  
}
```

Application.properties:

```
spring.application.name=Springlearner  
server.port=8083
```

OUTPUT:



Exersice 4 :

Spring Core – Load Country from Spring Configuration XML

Pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-
4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>3.5.3</version>
        <relativePath/> <!-- lookup parent from repository -->
    </parent>

    <groupId>com.cognizant</groupId>
    <artifactId>spring-learn</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <name>spring-learn</name>
    <description>Demo project for Spring Boot</description>

    <properties>
        <java.version>21</java.version>
    </properties>

    <dependencies>
        <!-- Spring Boot Web Starter -->
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
        </dependency>

        <!-- Spring Boot DevTools -->
```

```

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-devtools</artifactId>
  <scope>runtime</scope>
  <optional>true</optional>
</dependency>

<!-- SLF4J Logging -->
<dependency>
  <groupId>org.slf4j</groupId>
  <artifactId>slf4j-api</artifactId>
  <version>2.0.13</version>
</dependency>

<!-- Testing -->
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-test</artifactId>
  <scope>test</scope>
</dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-
plugin</artifactId>
    </plugin>
  </plugins>
</build>

</project>

```

Java class:

Package : controller

Class :CountryController

CountryController.java

```

package com.cognizant.spring_learn.controller;

import com.cognizant.spring_learn.model.Country; import
org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext; import
org.springframework.web.bind.annotation.RequestMapping; import
org.springframework.web.bind.annotation.RestController;

@RestController public class
CountryController {

    @RequestMapping("/country")
    public Country getCountryIndia() {
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = (Country) context.getBean("in");    return country;
    } }

```

Package : model

Class : Country

Country.java:

```

package com.cognizant.spring_learn.model;

import org.slf4j.Logger; import
org.slf4j.LoggerFactory;

public class Country {    private static final Logger LOGGER =
LoggerFactory.getLogger(Country.class);

    private String code;
    private String name;

    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() {

```

```

        LOGGER.debug("Inside getCode");
return code;
    }

    public void setCode(String code) {
LOGGER.debug("Inside setCode");    this.code
= code;
    }

    public String getName() {
        LOGGER.debug("Inside getName");
return name;
    }

    public void setName(String name) {
LOGGER.debug("Inside setName");    this.name
= name;
    }

    @Override    public String toString() {    return "Country
[code=" + code + ", name=" + name + "];    }
}

```

SpringLearnApplication.java

```

package com.cognizant.spring_learn;

import com.cognizant.spring_learn.model.Country; import
org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext;

public class SpringLearnApplication {

    public static void main(String[] args) {
System.out.println("START");    displayCountry();
        System.out.println("END");
    }

    public static void displayCountry() {

```



```
ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
Country country = (Country) context.getBean("in");

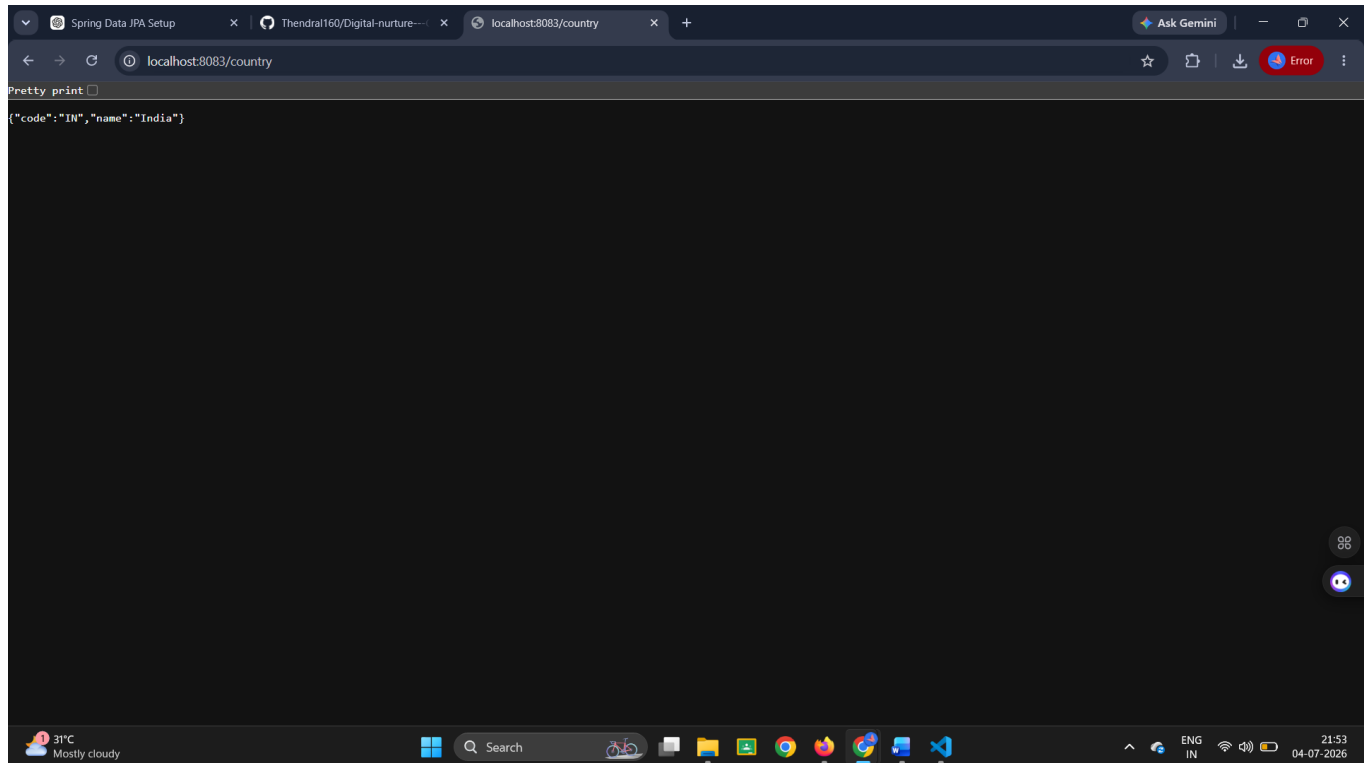
System.out.println("Country: " + country.toString());
} }
```

Country.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="in" class="com.cognizant.spring_learn.model.Country">
        <property name="code" value="IN"/>
        <property name="name" value="India"/>
    </bean> </beans>
```

OUTPUT:



Excercise 5: REST - Get country based on country code

Country.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="countryList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
```

```

    <bean class="com.cognizant.spring_learn.model.Country">
      <property name="code" value="IN"/>
      <property name="name" value="India"/>
    </bean>
    <bean class="com.cognizant.spring_learn.model.Country">
      <property name="code" value="US"/>
      <property name="name" value="United States"/>
    </bean>
    <bean class="com.cognizant.spring_learn.model.Country">
      <property name="code" value="JP"/>
      <property name="name" value="Japan"/>
    </bean>
  </list>
</constructor-arg>
</bean>
</beans>

```

CountryService.java:

```
package com.cognizant.spring_learn.service;
```

```
import com.cognizant.spring_learn.model.Country; import
org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext; import
org.springframework.stereotype.Service;
```

```
import java.util.List;
```

```
@Service
```

```
public class CountryService {
```

```
    public Country getCountry(String code) {
```

```
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
```

```
        List<Country> countryList = context.getBean("countryList", List.class);
```

```
        return countryList.stream()
```

```
            .filter(country -> country.getCode().equalsIgnoreCase(code))
```

```
            .findFirst()
```

```
            .orElse(null);
```

```
    } }
```

CountryController.java:

```
package com.cognizant.spring_learn.controller;

import com.cognizant.spring_learn.model.Country; import
com.cognizant.spring_learn.service.CountryService; import org.slf4j.Logger;
import org.slf4j.LoggerFactory; import
org.springframework.beans.factory.annotation.Autowired; import
org.springframework.context.ApplicationContext; import
org.springframework.context.support.ClassPathXmlApplicationContext; import
org.springframework.web.bind.annotation.*;

@RestController public class
CountryController {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryController.class);

    @Autowired    private CountryService
countryService;

    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("START");
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = context.getBean("in", Country.class);
        LOGGER.info("END");    return country;
    }

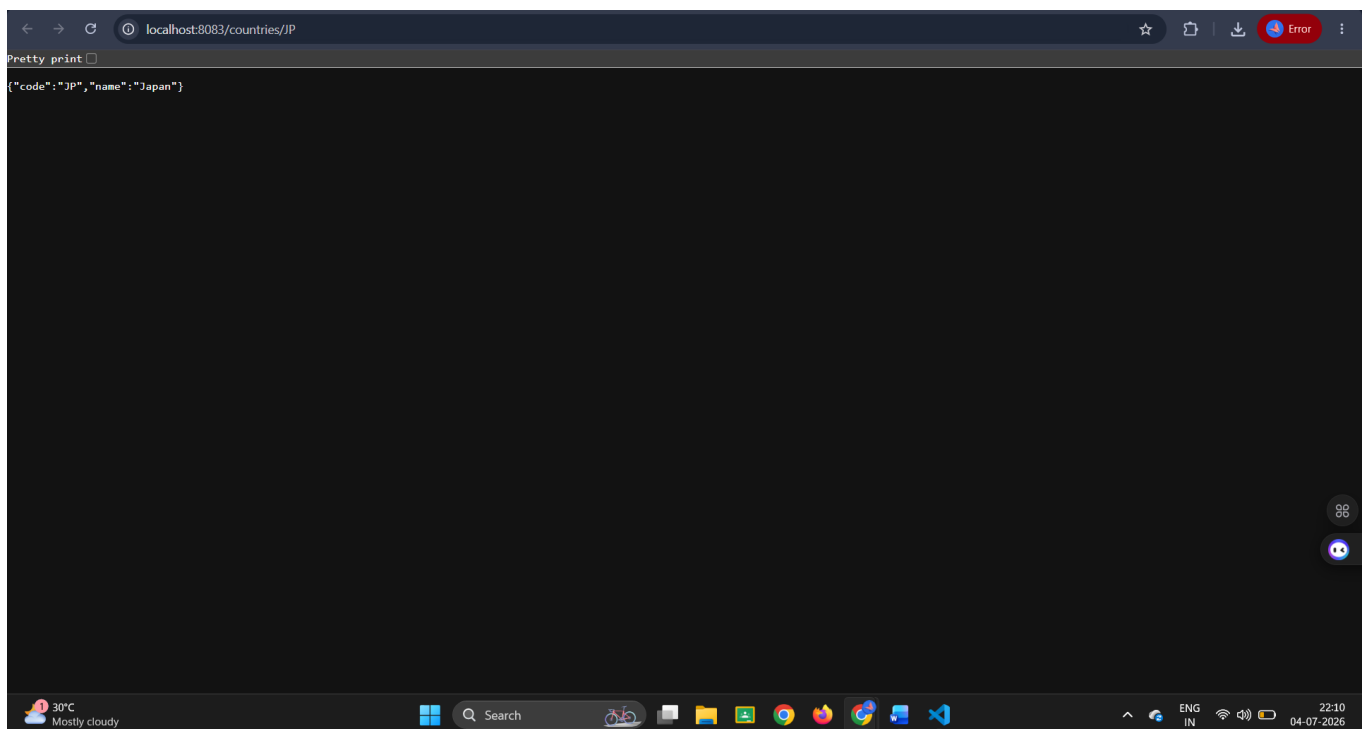
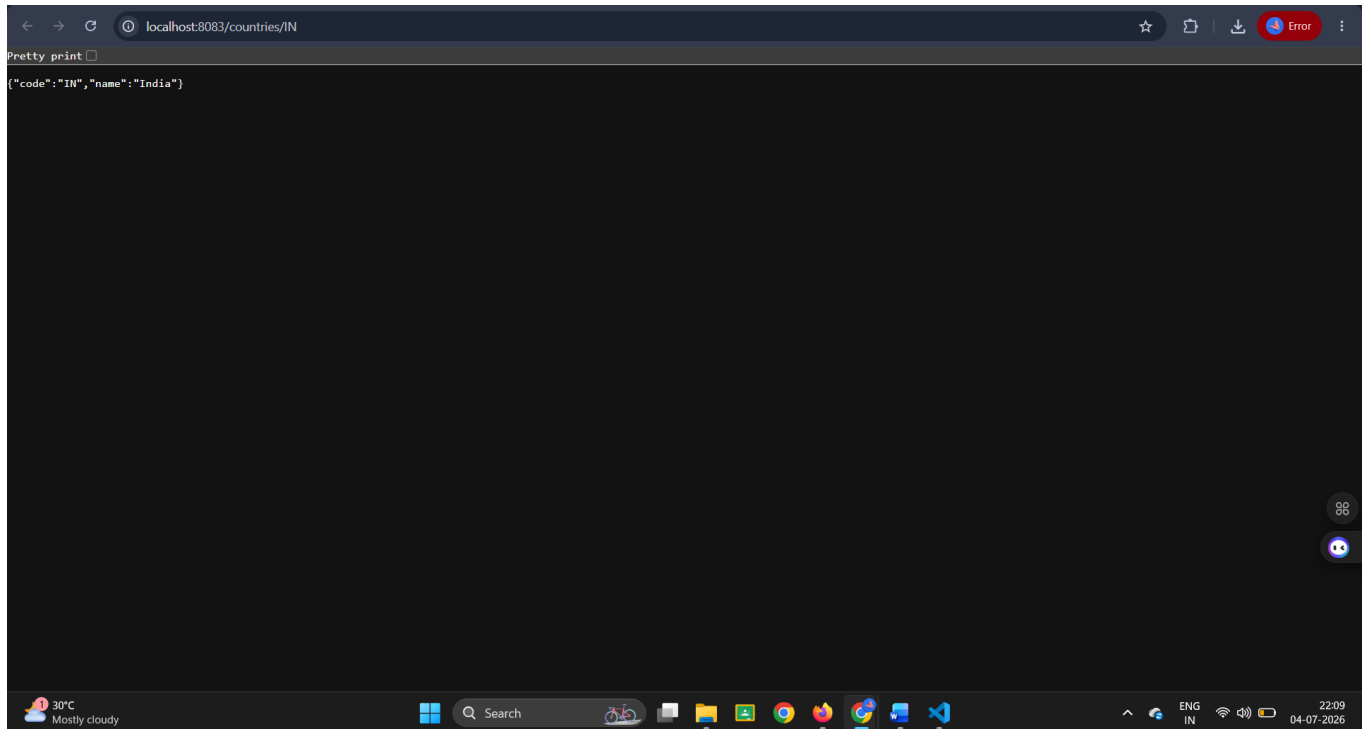
    @GetMapping("/countries/{code}")
    public Country getCountry(@PathVariable String code) {
        LOGGER.info("START");
        Country country = countryService.getCountry(code);
        LOGGER.info("END");    return country;    } }.
}
```

Output:

```
src > main > java > com > cognizant > spring_ > learn > J HelloController.java > HelloController
1 nackase.com.cognizant.spring_learn
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
:: Spring Boot :: (v4.1.0)
2026-07-04T22:06:05.929+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] c.c.s.learn.SpringLearnApplication : Starting S
pringLearnApplication using Java 26.0.1 with PID 7036 (c:\Users\thend\Downloads\spring-learn-ex5\target\classes started by thend in
C:\Users\thend\Downloads\spring-learn-ex5)
2026-07-04T22:06:05.933+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] c.c.s.learn.SpringLearnApplication : No active
profile set, falling back to 1 default profile: "default"
2026-07-04T22:06:06.000+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools p
roperty defaults active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-04T22:06:06.001+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additi
onal web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-04T22:06:06.770+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat ini
tialized with port 8083 (http)
2026-07-04T22:06:06.784+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] o.apache.catalina.core.StandardService : Starting s
ervice [Tomcat]
2026-07-04T22:06:06.784+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting s
ervlet engine: [Apache Tomcat/11.0.22]
2026-07-04T22:06:06.826+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebAp
plicationContext: initialization completed in 823 ms
2026-07-04T22:06:07.212+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat sta
rted on port 8083 (http) with context path '/'
2026-07-04T22:06:07.216+05:30 INFO 7036 --- [SpringLearner] [ restartedMain] c.c.s.learn.SpringLearnApplication : Started Sp
ringLearnApplication in 1.724 seconds (process running for 2.122)
2026-07-04T22:08:17.135+05:30 INFO 7036 --- [SpringLearner] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializi
ng Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T22:08:17.136+05:30 INFO 7036 --- [SpringLearner] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializi
ng Servlet 'dispatcherServlet'
2026-07-04T22:06:17.139+05:30 INFO 7036 --- [SpringLearner] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed
initialization in 1 ms
```

In browser:

```
localhost:8083/countries/US
Pretty print
{"code": "US", "name": "United States"}
```



Exersice 6: Create authentication service that returns JWT

Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
    <relativePath/>
  </parent>

  <groupId>com.cognizant</groupId>
  <artifactId>jwt-auth</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>jwt-auth</name>
  <description>JWT Auth Project</description>

  <properties>
    <java.version>24</java.version>
  </properties>

  <dependencies>
    <!-- Spring Boot Web and Security -->
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
```

```
    <artifactId>spring-boot-starter-security</artifactId>
</dependency>
```

```
<!-- JSON Web Token - JJWT -->
```

```
<dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-api</artifactId>
    <version>0.11.5</version>
```

```
</dependency>
```

```
<dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-impl</artifactId>
    <version>0.11.5</version>
    <scope>runtime</scope>
```

```
</dependency>
```

```
<dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-jackson</artifactId>
    <version>0.11.5</version>
    <scope>runtime</scope>
```

```
</dependency>
```

```
<!-- Servlet API -->
```

```
<dependency>
    <groupId>jakarta.servlet</groupId>
    <artifactId>jakarta.servlet-api</artifactId>
    <version>6.0.0</version>
    <scope>provided</scope>
```

```
</dependency>
```

```
<!-- Development and Testing -->
```

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
```



```

        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>
    <dependency>
        <groupId>org.springframework.security</groupId>
        <artifactId>spring-security-test</artifactId>
        <scope>test</scope>
    </dependency>
</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>
</project>

```

- Create package named jwt_auth and config,controller
- Path src/main/java/com/cognizant/jwt_auth/controller, src/main/java/com/cognizant/jwt_auth/config.
- Then create a class named AuthenticationController.java in src/main/java/com/cognizant/jwt_auth/controller/AuthenticationController.java.

AuthenticationController.java:

```
package com.cognizant.jwt_auth.controller;
```

```
import io.jsonwebtoken.Jwts; import
io.jsonwebtoken.SignatureAlgorithm; import
io.jsonwebtoken.security.Keys; import
org.springframework.web.bind.annotation.GetMapping; import
org.springframework.web.bind.annotation.RestController; import
jakarta.servlet.http.HttpServletRequest; import java.security.Key;
import java.util.Base64; import java.util.Date; import
java.util.HashMap; import java.util.Map;
```

```
@RestController public class
AuthenticationController {
```

```
    private final Key secretKey =
Keys.hmacShaKeyFor("ThisIsASecretKeyThatIsAtLeast32Chars!".getBytes());
    @GetMapping("/authenticate")
    public Map<String, String> authenticate(HttpServletRequest request) {
String authHeader = request.getHeader("Authorization");
```

```
        if (authHeader == null || !authHeader.startsWith("Basic ")) {            throw
new RuntimeException("Missing or invalid Authorization header");
    }
```

```
        String base64Credentials = authHeader.substring("Basic ".length());
        String credentials = new
String(Base64.getDecoder().decode(base64Credentials));
        String[] values = credentials.split(":", 2);
```

```
        String username = values[0];
        String password = values[1];
```

```
        if (!username.equals("user") || !password.equals("pwd")) {
throw new RuntimeException("Invalid credentials");
    }
```

```
        String token = Jwts.builder()
            .setSubject(username)
```

```

        .setIssuedAt(new Date())
        .setExpiration(new Date(System.currentTimeMillis() + 600000)) // 10
minutes
        .signWith(secretKey, SignatureAlgorithm.HS256)
        .compact();

    Map<String, String> response = new HashMap<>();
    response.put("token", token);    return response;
    } }

```

SecurityConfig.java:

```

package com.cognizant.jwt_auth.config;

import org.springframework.context.annotation.Bean; import
org.springframework.context.annotation.Configuration; import
org.springframework.security.config.annotation.web.builders.HttpSecurity; import
org.springframework.security.config.annotation.web.configuration.EnableWebSecu
rity;
import org.springframework.security.core.userdetails.User; import
org.springframework.security.core.userdetails.UserDetails; import
org.springframework.security.core.userdetails.UserDetailsService; import
org.springframework.security.provisioning.InMemoryUserDetailsManager; import
org.springframework.security.web.SecurityFilterChain;

@Configuration
@EnableWebSecurity public
class SecurityConfig {

    @Bean
    public UserDetailsService userDetailsService() {
        UserDetails user = User
            .withUsername("user")
            .password("{noop}pwd") // {noop} means no encoding
            .roles("USER")
            .build();
    }
}

```

```

        return new InMemoryUserDetailsManager(user);
    }

    @Bean    public SecurityFilterChain filterChain(HttpSecurity http) throws
Exception {    http.csrf(csrf -> csrf.disable())
                .authorizeHttpRequests(auth -> auth
                    .requestMatchers("/authenticate").authenticated()
                .anyRequest().permitAll()
            )
                .httpBasic(); // enables basic auth
    return http.build();
    } }

```

Application.properties:

```

spring.application.name=jwt-auth
server.port=8080

```

JwtAuthApplication.java:

```

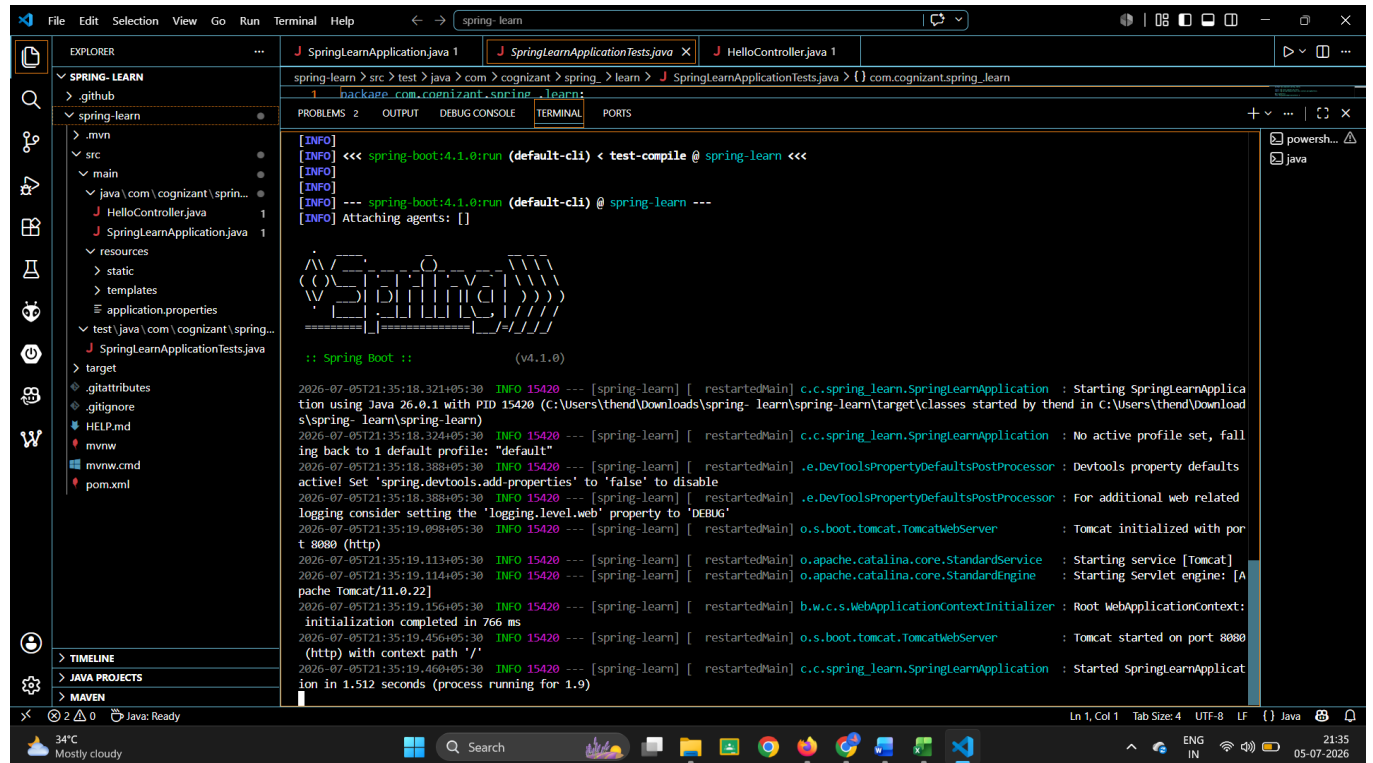
package com.cognizant.jwt_auth;

import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication public class
JwtAuthApplication {    public static
void main(String[] args) {
    SpringApplication.run(JwtAuthApplication.class, args);
    } }

```

Output:



The terminal token generation:

```
PS D:\Placement\CTS\Projects\jwt-auth> jwt-auth -u user:pwd http://localhost:8080/authenticate
{"token": "eyJhbGciOiJIUzI1NiIsInR5eSI6ImlwIj09IiwiaWF0IjoxNzUyMTQ9LCJleHAiOjE3NTI1NDUyNjZ9.eyJkIj09IiwiaWF0IjoxNzUyMTQ9LCJleHAiOjE3NTI1NDUyNjZ9.eyJkIj09IiwiaWF0IjoxNzUyMTQ9LCJleHAiOjE3NTI1NDUyNjZ9"}
PS D:\Placement\CTS\Projects\jwt-auth> jwt-auth -t http://localhost:8080/authenticate
```